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Domain : AI/ML

Q1. (Preparing Features)

```
▶ X = df.drop("price", axis=1)
  y = df["price"]
  X.shape
  y.shape
```

```
... (17966,)
```

Q2. (One-Hot Encoding)

```
▶ X = pd.DataFrame({  
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],  
    'Gender': ['F', 'M', 'M', 'M'],  
    'City': ['Delhi', 'Mumbai', 'Delhi', 'Chennai'],  
    'Age': [25, 30, 35, 40]  
})  
X = pd.get_dummies(X).astype(int)  
print(X.head())
```

```
...   Age  Name_Alice  Name_Bob  Name_Charlie  Name_David  Gender_F  Gender_M  \  
0    25           1           0              0           0           1           0  
1    30           0           1              0           0           0           1  
2    35           0           0              1           0           0           1  
3    40           0           0              0           1           0           1  
  
      City_Chennai  City_Delhi  City_Mumbai  
0                0            1            0  
1                0            0            1  
2                0            1            0  
3                1            0            0
```

Q3. (Feature Scaling)

```
▶ x = pd.DataFrame({  
    'year': [2015, 2016, 2017, 2018, 2019],  
    'mileage': [15000, 12000, 30000, 25000, 18000],  
    'tax': [150, 180, 200, 210, 190],  
    'mpg': [55.4, 60.1, 58.9, 57.2, 59.0],  
    'engineSize': [1.6, 2.0, 1.8, 2.2, 2.0]  
})  
scaler = StandardScaler()  
scaled = scaler.fit_transform(x[['year', 'mileage', 'tax', 'mpg', 'engineSize']])  
  
scaled_df = pd.DataFrame(scaled, columns=['year', 'mileage', 'tax', 'mpg', 'engineSize'])  
  
print(scaled_df.head())
```

```
...      year  mileage      tax      mpg  engineSize  
0 -1.414214 -0.757228 -1.748315 -1.652404 -1.568929  
1 -0.707107 -1.211565 -0.291386  1.202853  0.392232  
2  0.000000  1.514456  0.679900  0.473851 -0.588348  
3  0.707107  0.757228  1.165543 -0.558901  1.372813  
4  1.414214 -0.302891  0.194257  0.534601  0.392232
```

Q4. (Train-Test Split)

```
▶ X = pd.DataFrame({  
    'year': [2015, 2016, 2017, 2018, 2019],  
    'mileage': [15000, 12000, 30000, 25000, 18000],  
    'tax': [150, 180, 200, 210, 190],  
    'mpg': [55.4, 60.1, 58.9, 57.2, 59.0],  
    'engineSize': [1.6, 2.0, 1.8, 2.2, 2.0]  
})  
y = [0, 1, 0, 1, 0]  
  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33, random_state=42)  
  
print(X_train.shape, X_test.shape, len(y_train), len(y_test))
```

```
... (3, 5) (2, 5) 3 2
```

Q5. (Building Linear Regression Model)

```
▶ X = pd.DataFrame({
    'year': [2015, 2016, 2017, 2018, 2019],
    'mileage': [15000, 12000, 30000, 25000, 18000],
    'tax': [150, 180, 200, 210, 190],
    'mpg': [55.4, 60.1, 58.9, 57.2, 59.0],
    'engineSize': [1.6, 2.0, 1.8, 2.2, 2.0]
})
y = [0, 1, 0, 1, 0]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)

... Intercept: -8.391274378902668
Coefficients: [ 0.002333 -0.00012457  0.03732782 -0.00074659  0.00065324]
```

Q6. (Making Predictions)

```
▶ y_pred = model.predict(X_test)

print("Predicted:", y_pred[:10])
print("Actual:", y_test[:10])
```

```
... Predicted: [1.49263498 1.12630356]
Actual: [1, 0]
```

Q7. (Model Evaluation using R2 Score)

```
▶ from sklearn.metrics import r2_score  
  
r2 = r2_score(y_test, y_pred)  
print("R2 Score:", r2)
```

```
... R2 Score: -2.0224978533391478
```


Q8. (Saving the Model)

```
import joblib
joblib.dump(model, "LR_ford_car.pkl")
joblib.dump(scaler, "scaler.pkl")
joblib.dump(X.columns.tolist(), "columns.pkl")
```

```
['columns.pkl']
```

Optional Task:

```
df = pd.read_csv("fordcar.csv")

X = df[['year', 'mileage', 'tax', 'mpg', 'engineSize']]
y = df['price']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.33, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
print("R2 Score:", r2_score(y_test, y_pred))
```

```
... Intercept: 12279.27131131649
Coefficients: [ 2491.75093619 -1161.0129724      6.62091728 -828.57935272
 2321.50325537]
R2 Score: 0.7246904411329367
```

❖ <https://github.com/surajjagtap08062007-sj/surajpython.git>